



Project Procurement Management

IT/Software Project Management

What is project procurement management?

- It is part of project management that includes the processes required **to acquire goods and services** to attain project scope from **outside** the performing organization.
- Other terms include **purchasing** and **outsourcing**
- The **global IT outsourcing market (total IT services outsourcing)** is expected to generate around **US \$574 billion in 2025** according to industry forecast data (growing steadily from previous years).

Why we outsource?

- To reduce both fixed and recurrent costs
- To allow the client organization to focus on its core business
- To access skills and technologies
- To provide flexibility
- To increase accountability

Project procurement management processes

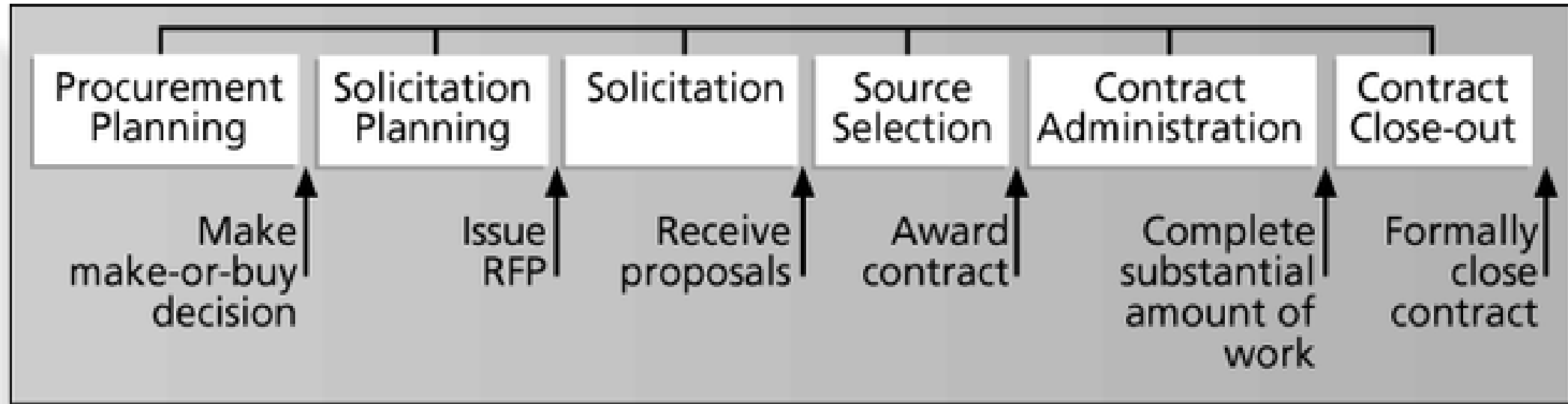
- Processes include:

1. **Procurement Planning:** determining **what** to procure, **when**, and **how**.
2. **Solicitation Planning:** describing requirements for the products or services desired from the procurement and **identifying potential sources or sellers** (contractors, suppliers, or providers who provide goods and services to other organizations).
3. **Solicitation:** obtaining information, quotes, bids, offers, or proposals from sellers, as appropriate.

...processes continued

4. **Source Selection:** choosing from among potential suppliers through a process of evaluating potential sellers and negotiating the contract.
5. **Contract Administration:** managing the relationship with the selected seller.
6. **Contract Closeout:** completing and settling each contract, including resolving any open items.

Project Procurement Management Processes and Key Outputs



Procurement Planning

- Identifying which project needs can best be met by using products or services outside the organization.
- If there is no need to buy any products or services from outside the organization, then there is **no need to perform any of the other procurement management processes.**

...Procurement Planning

- Procurement planning involves identifying which project needs can be best met by using products or services outside the organization.
- It includes deciding
 - Whether to procure ...?
 - How to procure
 - What to procure
 - How much to procure
 - When to procure

Procurement planning tools and techniques

- **Make-or-buy analysis:** determining whether a particular product or service should be made or performed inside the organization or purchased from someone else.

Often involves financial analysis

- Experts, both internal and external, can provide valuable inputs in procurement decisions

Make-or buy example

- Assume you can lease an item you need for a project for \$800/day. To purchase the item, the cost is \$12,000 plus a daily operational cost of \$400/day.
- How long will it take for the purchase cost to be the same as the lease cost?

Make-or buy ...solution

- ▣ Set up an equation so both options, purchase and lease, are equal.
- ▣ In this example, use the following equation. Let d be the number of days to use the item:

$$\$12,000 + \$400d = \$800d$$

Subtracting $\$400d$ from both sides, you get:

$$\$12,000 = \$400d$$

Dividing both sides by $\$400$, you get:

$$d = 30$$

- ▣ If you need the item for more than 30 days, it is more economical to purchase it.

Contracts

- A **contract** is a mutually binding agreement that obligates the seller to provide the specified products or services and obligates the buyer to pay for them.
- Contracts can clarify responsibilities and sharpen focus on key deliverables of a project.
- Because contracts are legally binding, there is more accountability for delivering the work as stated in the contract.
- A recent trend in outsourcing is the increasing size of contracts.

Types of contracts

- **Fixed-price** or **lump-sum**: involve a fixed total price for a well-defined product or service.
- **Cost-reimbursable**: involve payment to the seller for direct and indirect costs
- **Time and material contracts**: hybrid of both fixed-price and cost-reimbursable, often used by consultants
- **Unit price contracts**: require the buyer to pay the seller a predetermined amount per unit of service

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Procurement management plan

- Describes how the procurement processes will be managed, from developing documentation for making outside purchases or acquisitions to contract closure.
- Contents vary based on project needs.

Statement of Work (SOW)

- A **statement of work** is a description of the work required for the procurement.
- If a SOW is used as part of a contract to describe only the work required for that particular contract, it is called a **contract statement of work**.
- A SOW is a type of scope statement.
- A good SOW gives bidders a better understanding of the buyer's expectations.

Statement of Work (SOW) Template

- I. **Scope of Work:** Describe the work to be done to detail. Specify the hardware and software involved and the exact nature of the work.
- II. **Location of Work:** Describe where the work must be performed. Specify the location of hardware and software and where the people must perform the work
- III. **Period of Performance:** Specify when the work is expected to start and end, working hours, number of hours that can be billed per week, where the work must be performed, and related schedule information.
- IV. **Deliverables Schedule:** List specific deliverables, describe them in detail, and specify when they are due.
- V. **Applicable Standards:** Specify any company or industry-specific standards that are relevant to performing the work.
- VI. **Acceptance Criteria:** Describe how the buyer organization will determine if the work is acceptable.
- VII. **Special Requirements:** Specify any special requirements such as hardware or software certifications, minimum degree or experience level of personnel, travel requirements, and so on.

Plan contracting

- Involves preparing several documents needed for potential sellers to prepare their responses and determining the evaluation criteria for the contract award.
- **Request for Proposals (RFP):** Used to solicit proposals from prospective sellers.
 - A **proposal** is a document prepared by a seller when there are different approaches for meeting buyer needs.
- **Requests for Quotes (RFQ):** Used to solicit quotes or bids from prospective suppliers.
 - A **bid**, also called a tender or quote (short for quotation), is a document prepared by sellers providing pricing for standard items that have been clearly defined by the buyer.

Outline for a Request for Proposal (RFP)

- I. Purpose of RFP
- II. Organization's Background
- III. Basic Requirements
- IV. Hardware and Software Environment
- V. Description of RFP Process
- VI. Statement of Work and Schedule Information
- VII. Possible Appendices
 - A. Current System Overview
 - B. System Requirements
 - C. Volume and Size Data
 - D. Required Contents of Vendor's Response to RFP
 - E. Sample Contract

Evaluation criteria

- It's important to prepare some form of evaluation criteria, preferably before issuing a formal RFP or RFQ.
- Beware of proposals that look good on paper; be sure to evaluate factors, such as **past performance and management approach**.
- Can require a **technical presentation as part of a proposal**.

Requesting seller responses

- Deciding whom to ask to do the work, sending appropriate documentation to potential sellers, and obtaining proposals or bids.
- Organizations can advertise to procure goods and services in several ways:
 - Approaching the preferred vendor.
 - Approaching several potential vendors.
 - Advertising to anyone interested.
- A bidders' conference can help clarify the buyer's expectations.

Selecting sellers

- Also called source selection.
- Involves:
 - Evaluating proposals or bids from sellers.
 - Choosing the best one.
 - Negotiating the contract.
 - Awarding the contract.

Sample proposal evaluation sheet

		Proposal 1		Proposal 2		Proposal 3	
Criteria	Weight	Rating	Score	Rating	Score	Rating	Score
Technical Approach	30%						
Management Approach	30%						
Past Performance	20%						
Price	20%						
Total Score	100%						

Detailed criteria for selecting suppliers

Criteria	Possible Points	Supplier 1 Points	Supplier 2 Points	Supplier 3 Points
Project manager s educational background and experience	10	8	6	9
Project manager is PMP certified	5	5	0	5
Presentation on management approach	5	4	3	5
Organization s project management methodology	10	7	4	9
Total Score	30	24	13	28

Be careful in selecting suppliers and writing their contracts

- Many dot-com companies were created to meet potential market needs, but many went out of business, mainly due to poor business planning, lack of senior management operations experience, lack of leadership, and lack of visions.
- Check the stability of suppliers
- Even well-known suppliers can impede project success. Be sure to write and manage contracts well with all suppliers.

Seller selection process

- Organizations often do an initial evaluation of all proposals and bids and then develop a short list of potential sellers for further evaluation.
- Sellers on the short list often prepare a best and final offer
- Final output is a contract signed by the buyer and the selected seller.

Administering the contract

- Contract administration ensures that the seller's performance meets contractual requirements
- Contracts are **legal relationships**, so it is important that **legal and contracting professionals** be involved in writing and administering contracts
- Many project managers ignore contractual issues, which can result in serious problems

Suggestions on change control for contracts

- Changes to any part of the project need to be reviewed, approved, and documented by the same people in the same way that the original part of the plan was approved
- Evaluation of any change should include an impact analysis. How will the change affect the scope, time, cost, and quality of the goods or services being provided?
- Changes must be documented in writing. Project team members should also document all important meetings and telephone calls

Suggestions on change control for contracts (cont...)

- Project managers and teams should stay closely involved to make sure the new system will meet business needs and work in an operational environment.
- Have backup plans.
- Use tools and techniques, such as a contract change control system, buyer-conducted performance reviews, inspections and audits, and so on.

Closing the contract

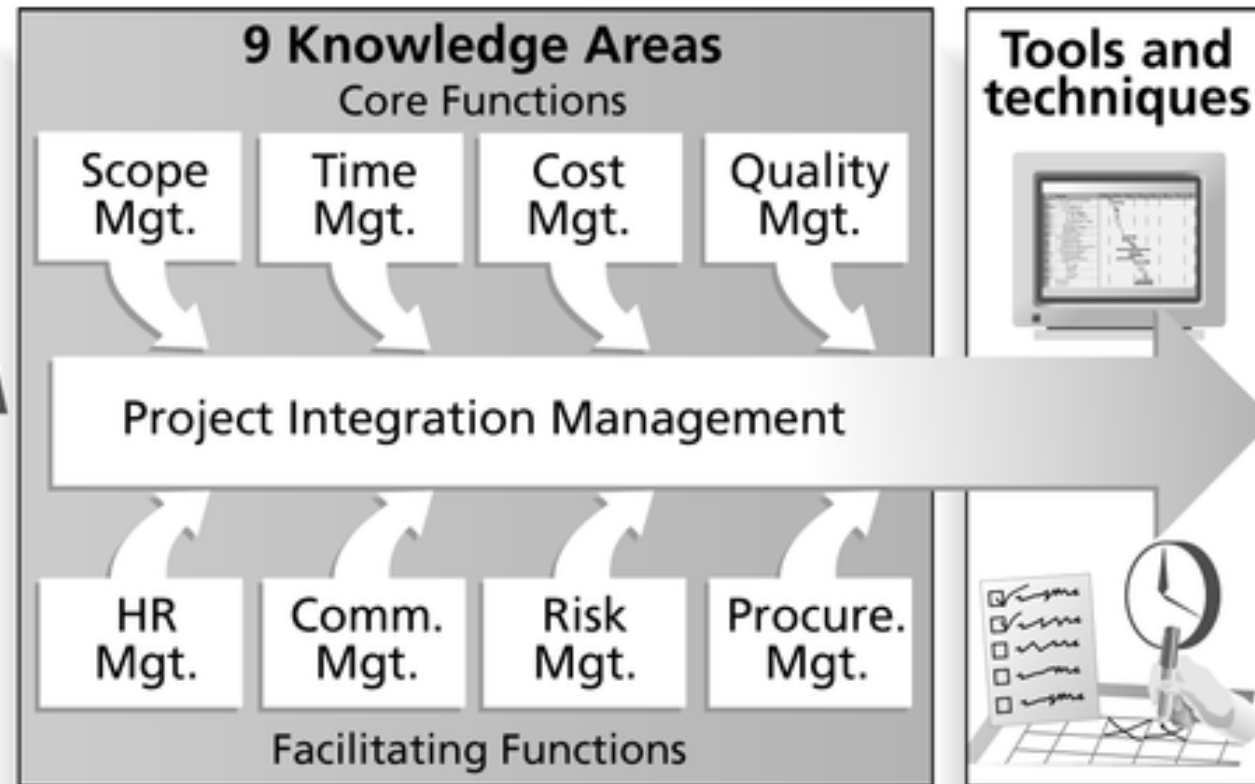
- Involves completing and settling contracts and resolving any open items.
 - The project team should:
 - Determine if all work was completed correctly and satisfactorily.
 - Update records to reflect final results.
 - Archive information for future use.
 - The contract itself should include requirements for formal acceptance and closure.
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Using software to assist in project procurement management

- Word processing software helps in writing proposals and contracts, spreadsheets help in evaluating suppliers, databases help track suppliers, and presentation software aids in presenting procurement-related information
- E-procurement software does many procurement functions electronically.
- Organizations also use other Internet tools to help find information on suppliers or auction goods and services



Stakeholders'
needs and
expectations



Project
Success

